COMMODITIES PRICES IN GAZA

THE PRICES BEFORE AND AFTER THE WAR

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Data Source: Palestinian Central Bureau of Statistics (PCBS)



1.Problem Statement

This project focuses on the dramatic changes in the prices of essential goods and services before the war and throughout its extended months in the Gaza Strip. Its significance lies in shedding light on the immense suffering endured by the population due to the war and the daily living conditions they face. Through this project, we attempt to answer the underlying causes behind this drastic surge in prices. The importance of this data goes beyond mere numbers — behind every price change are people struggling to secure their basic needs amid a harsh reality that affects every segment of society in the Strip.

2.Data Source & Scope

Metric	Value
Number of commodities analyzed	46
Time period	Sep 2023 – Aug 2026
Data type	Monthly prices + % change
Source	Palestinian Central Bureau of Statistics (PCBS)

The original data from the source was preserved without any modifications, to maintain the credibility of the analysis.

Methodology (MEAL Framework)

Metrics:

- Shock Change % — immediate price jump after Oct 7, 2023
- Cumulative Change % — total change over the full period
- Peak Price — highest price reached by each commodity

Entities:

- Commodity (individual item)
- Commodity Category (Water / Fuel / Food / Non-Essential)

Attributes:

- Unit of measurement
- Price before/after war

Logs:

- A single dataset of 46 commodities with monthly documented prices from one official source

4.Key Findings

Finding #1 — Transportation: The Sharpest Shock

Commodity	Immediate Shock
Transport (North → Central Gaza)	+2400%
Transport (North → South Gaza)	+1775%
Diesel	+208%
Gasoline	+199%

The data shows that transportation prices recorded the highest increase, driven by the shortage of vehicle spare parts and severe fuel scarcity resulting from the blockade imposed on the Strip, along with a sharp decline in the number of functioning vehicles — many of which were destroyed during the war. Notably, the fuel crisis itself (diesel and gasoline) directly fed into transportation costs, forming an interconnected chain of crises.

Finding #2 — The Silent Water Crisis

Commodity	Cumulative Change
Trucked water (250L)	+365%
Trucked water (500L)	+407%
Trucked water (1000L)	+522%

The focus on the water crisis stems from the direct risks it poses to residents' lives. The destruction of desalination plants, combined with the repeated displacement of residents from their homes, has severely limited access to water sources. In addition, the extreme surge in fuel prices further drove up the cost of water, as pumping and transporting it became increasingly expensive.

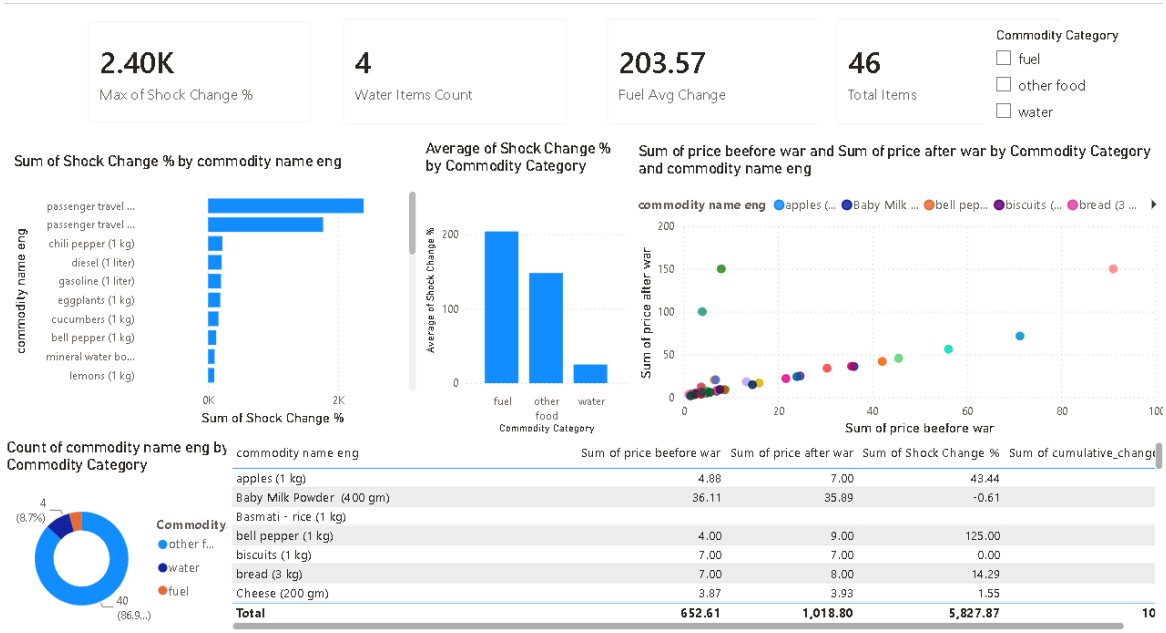
Finding #3 — Fuel Leads Category-Level Change

Commodity	Average Change
Fuel	Highest
Other Food	Moderate
Water	Lower relative average (still very high)

The table shows the categories most affected by price increases. It is worth noting, however, that water shows a comparatively lower impact than fuel — a finding that carries significant implications, as it reflects how the surge in fuel prices ripples across nearly every aspect of daily life in the Strip.

5.Interactive Power BI Dashboard

- ☐ 4 KPI Cards — max shock %, water items count, average fuel change, total items analyzed
- ☐ Bar Chart — top commodities by price shock
- ☐ Bar Chart — average change by category
- ☐ Scatter Chart — price before vs. after the war
- ☐ Donut Chart — distribution by category
- ☐ Detailed Table — full figures per commodity
- ☐ Interactive Slicer — filter by category



Data Limitations

Commodity	Price Before War
Egyptian rice (1 kg)	Not recorded
Basmati rice (1 kg)	Not recorded
Egyptian flour (25 kg)	Not recorded
Egyptian flour (50 kg)	Not recorded

A small number of commodities — Egyptian rice, Basmati rice, and Egyptian flour (25kg and 50kg) — had no recorded price before the war. These appear to be substitute imports that entered the local market only after the crisis began, replacing local varieties that became unavailable. As a result, the percentage change could not be calculated for these specific items. This is noted as a methodological limitation rather than a flaw in the analysis.

Recommendations

Based on these findings, the following recommendations are proposed for consideration by international humanitarian organizations:

1. Facilitate the entry of essential goods into Gaza through coordinated humanitarian channels, prioritizing items shown to have the highest price shock — most notably fuel and basic food staples.

2. Advocate for the entry of fuel and vehicle spare parts, given their direct role in driving up transportation costs and their cascading effect on nearly every other sector of daily life.

3. Support the rehabilitation of water infrastructure, including the supply of well-drilling materials and the reopening of desalination plants, to address the sharp and sustained rise in water prices and its direct impact on public health.

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